

Anupa Supul

Intern Software Engineer

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Summary

Computer Science undergraduate with hands-on experience in software development, focusing on full-stack and backend systems. Strong foundation in object-oriented programming, database management, REST API development, and modern web technologies. Also experienced in deep learning research on rice leaf disease detection, including CNN-based model evaluation and application development. Passionate about building scalable applications, learning new technologies, and contributing effectively in team environments.

Education

University of Ruhuna - Bachelor of Computer Science

September 2023–Present

St. Thomas' Boys' College, Matara

- G.C.E. Advanced Level – Physical Science
- Combined Mathematics – A, Physics – C, Chemistry – B, Z-score: 1.4847

Technical Skills

Programming Languages: Java, JavaScript

Backend: Spring Boot, Express.js

Frontend: React.js, Next.js, Tailwind CSS

Databases: MySQL, MongoDB

AI/ML: TensorFlow, Keras, CNN, VGG16, VGG19

Tools: Git, GitHub, Postman, Trello, Figma

Projects

MediCare — Hospital Management System(Final Year Group Project)

October 2025–Present

- **Technologies:** Spring Boot, React.js, TailwindCSS, JWT Authentication, MySQL, Google Gemini API
- Developed a scalable hospital management system to manage patients, appointments, and staff workflows.
- Designed and implemented backend services using **Spring Boot with microservices architecture**, enabling service separation and scalability.
- Implemented secure authentication and authorization using JWT.
- Built responsive frontend interfaces using React.js and Tailwind CSS.
- Integrated Google Gemini API for AI-powered chatbot assistance.
- Integrated a pre-trained ECG machine learning model and implemented rule-based logic for automated health data analysis and intelligent alerts.
- Used MySQL for secure data management and GitHub/Trello for collaboration and project tracking.

Symphony Institute—Web Application

November 2025–Present

- **Technologies:** React.js, Tailwind CSS, Vite, JavaScript, Figma
- Designed and developed a responsive institute/portfolio website with modern UI components and clean layout structure.
- Developed the Symphony Institute portfolio website by implementing UI/UX designs provided by a designer, ensuring a fully responsive layout optimized for web, tab and mobile screens
- Developed and implemented a 6-page responsive website using clean, reusable React components and best practices.

Maths Class Management System

Combined Mathematics – Mr. Gayan Gammanavidana

September 2025–Present

- **Tech Stack:** MERN Stack (MongoDB, Express.js, React.js, Node.js)
- Developed a full-stack class management system for a Combined Mathematics instructor.
- Implemented QR code generation and scanning to automate student attendance and payment verification.

- Built modules for attendance tracking, payment management, and student records.
- Designed responsive user interfaces using React.js for both administrative and operational workflows.
- Collaborated directly with the client to gather requirements and refine system features.
- Currently in the final deployment phase.

Rice Plant Disease Detection(Research based)—WebApplication

December 2025 — *Present*

- **Technologies:** React.js,TailwindCSS,SpringBoot,MySQL, REST APIs,TensorFlow/Keras, VGG19
- Developed a research-oriented rice leaf disease detection web application that captures leaf images through a camera and automatically classifies the disease using deep learning. Built an end-to-end system including a responsive frontend and a backend API to handle image uploading, preprocessing, and inference. Trained and evaluated multiple CNN-based models (VGG19) to improve detection accuracy and reliability. Implemented the solution as a practical tool to support fast and accessible disease identification for agricultural use cases .

Rice Leaf Disease Detection(Research)

November2024 – *Present*

- Conducted deep learning research to classify rice leaves as healthy or diseased using image-based detection.
- Created and curated the data set by collecting rice leaf images data for training and testing.
- Applied data preprocessing and augmentation (resize, normalization, rotation, brightness adjustments) to improve model generalization.
- Implemented and compared multiple architectures including VGG16 ,VGG19 ,custom CNN and CNN+SVM for performance analysis.
- Trained and evaluated a CNN model using a **70/20/10** train/validation/test split, achieving **95.27% validation accuracy** and **92.62% test accuracy** for brown spot detection.

Research publications

• Identification Browns spot Leaf Disease Using Machine Learning

11th November 2025

Student Symposium - University of Sabaragamuwa

Published research paper at APSURS 2025 (4th Applied Sciences Undergraduate research Symposium), Sabaragamuwa University Sri Lanka

• Machine Learning and Deep Learning-Based Approaches for Rice Leaf Disease Detection in Sri Lanka

28th February 2026

Ritscon 2026 - University of Ruhuna

Research paper accepted / to be presented at RITSCON 2026, focusing on rice disease identification using machine learning and deep learning techniques based on captured leaf images.

Upcoming publications and Projects

• Rice Plant Leaf Disease Image Dataset

Planned release: March 2026

Currently developing and preparing a rice leaf image dataset captured under real-world conditions for machine learning and deep learning research. The dataset is planned for public release to support academic and research use.

Volunteering & Community Involvement

• **Organizing Committee Member—Public Relations Team Leader** |RushCoder3.0Hackathon,University of Ruhuna

• **Member** |Sasnaka sansadaya

References

Mr. L. G. G. Chathuranga Lecturer, Department of Computer Science University of Ruhuna
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